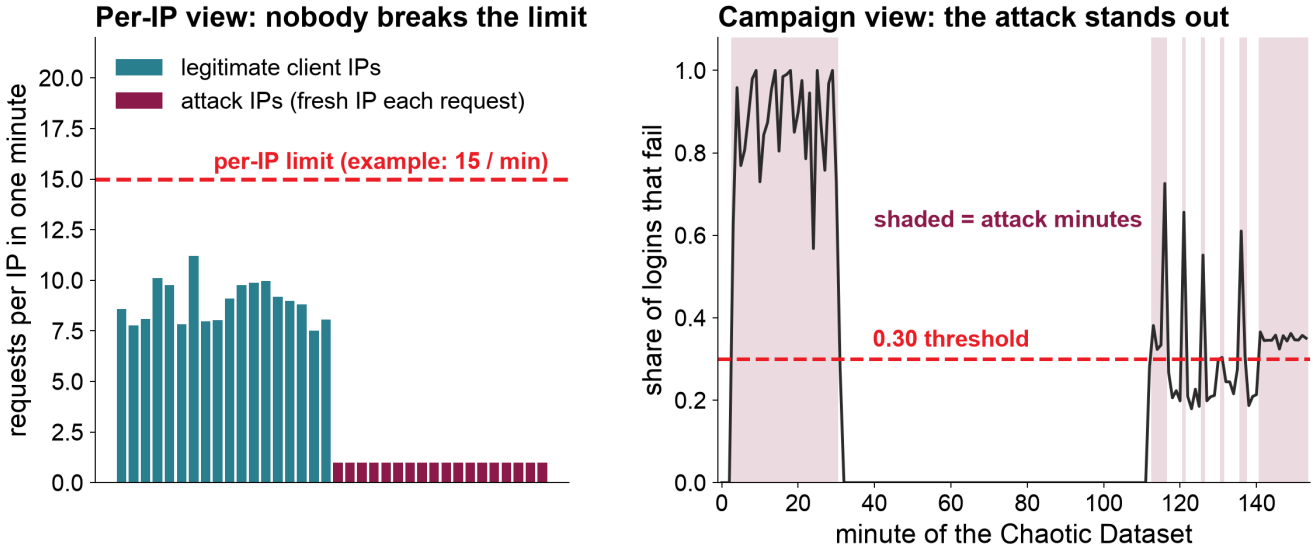
Campaign-Level Detection of Low-and-Slow Password Spraying at API Login Endpoints Using a Temporal Convolutional Network

Chia U-Meng Adrian (Xie Youming) · MTCS-MRMC5

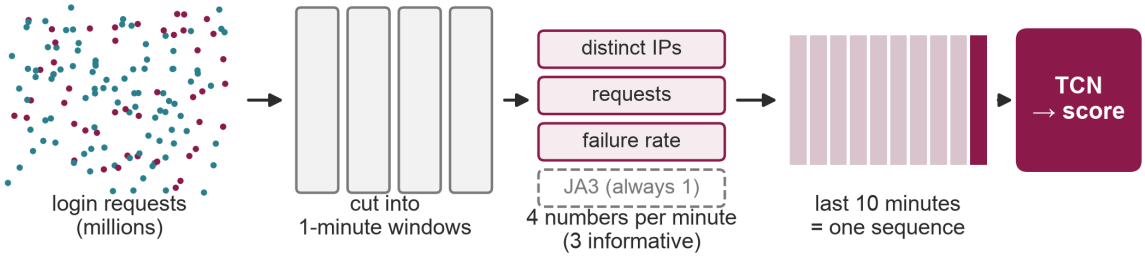
The problem

Attackers try one common password against many accounts (password spraying) and send every attempt from a different IP address, slowly. Every IP and every account stays under its rate limit or lockout threshold, so per-client defences see nothing.



The idea

Watch the crowd, not the person: summarise all login traffic in each minute (distinct IPs, requests, share of failed logins) and let a small Temporal Convolutional Network (TCN) score the last ten minutes once a minute.



Results (simulated testbed, two independent test datasets)

Measure	Large Dataset (233.4 h)	Chaotic Dataset (154 min)
Attack sequences found (recall)	1,992 of 1,992 (1.0000)	34 of 44 (0.7727)
False alarms	11 of 11,359 benign (≈ 1.4 per day)	4 of 101 benign
Precision · F1	0.9945 · 0.9972	0.8947 · 0.8293
Time per decision	0.65 ms mean (≈ 0.001% of one CPU core)	same model

Plot twist: in training, attacks used more IP addresses than normal traffic; in the Large Dataset they used fewer. The model still found every attack, because its scores follow the failure rate.

What this does not show

The traffic is simulated; the labels come from a rule (failure rate > 0.30, more than 5 IPs) that the model learns; TLS and header features and the throttling step were not tested; results come from a single training run. An attacker who knows the boundary could add successful logins to hide.